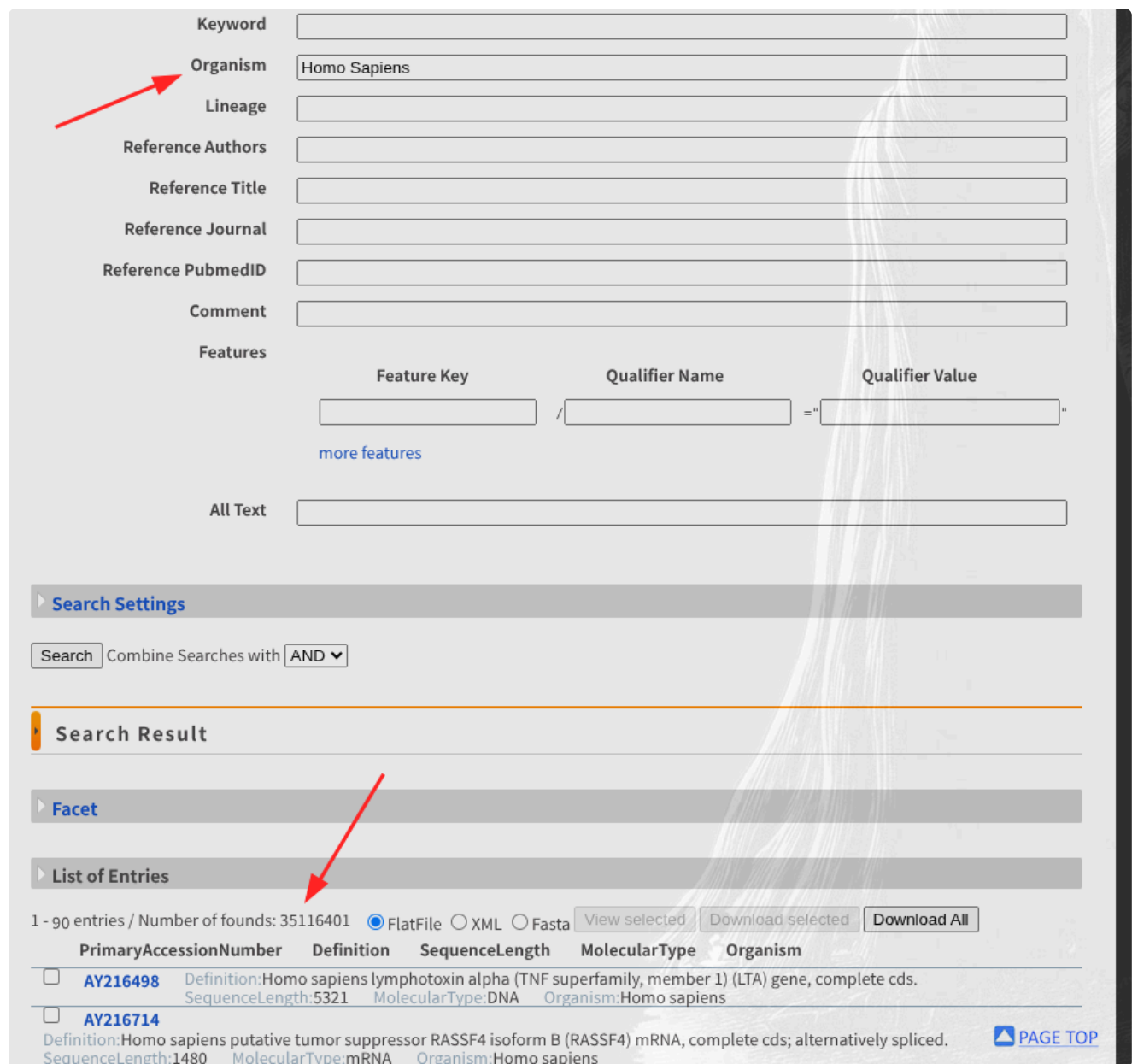


Solution - 2

Q1

DDJB

this link opens the DDJB advanced search for Homo sapiens



Keyword

Organism

Lineage

Reference Authors

Reference Title

Reference Journal

Reference PubmedID

Comment

Features

Feature Key	Qualifier Name	Qualifier Value
	/	= " "

[more features](#)

All Text

Search Settings

Search Combine Searches with

Search Result

Facet

List of Entries

1 - 90 entries / Number of founds: 35116401 ☒ FlatFile ☐ XML ☐ Fasta

PrimaryAccessionNumber	Definition	SequenceLength	MolecularType	Organism
☐ AY216498	Definition:Homo sapiens lymphotoxin alpha (TNF superfamily, member 1) (LTA) gene, complete cds.	SequenceLength:5321	MolecularType:DNA	Organism:Homo sapiens
☐ AY216714	Definition:Homo sapiens putative tumor suppressor RASSF4 isoform B (RASSF4) mRNA, complete cds; alternatively spliced.	SequenceLength:1480	MolecularType:mRNA	Organism:Homo sapiens

[PAGE TOP](#)

so DDBJ = 35,116,401

genbank

NIH National Library of Medicine
National Center for Biotechnology Information

Log in

Nucleotide Homo Sapiens[Organism] [Create alert](#) [Advanced](#) [Help](#)

Species: Animals (29,784,649), Plants (1), Fungi (1), Viruses (4,548), Customize ...

Molecule types: genomic DNA/RNA (16,937,4), mRNA (9,950,426), rRNA (788), Customize ...

Source databases: INSDC (GenBank) (29,256,885), RefSeq (424,468), Customize ...

Sequence Type

Summary 20 per page Sort by Default order

Items: 1 to 20 of 29841998

<< First < Prev Page 1 of 1492100 Next > Last >>

☐ [Homo sapiens N-sulfoglucosamine sulfohydrolase \(SGSH\), RefSeqGene on chromosome 17](#)
26,661 bp linear DNA
Accession: NG_008229.2 GI: 2310189229
[Protein](#) [PubMed](#) [Taxonomy](#)
[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Homo sapiens FASTK mitochondrial RNA regulator 3 \(FASTKD3\), transcript variant 3, non-coding RNA](#)
1,001 bp linear transcribed-RNA
Accession: NR_073608.2 GI: 1890392847
[PubMed](#) [Taxonomy](#)
[GenBank](#) [FASTA](#) [Graphics](#)

Filters: [Manage Filters](#)

Send to:

Results by taxon

Top Organisms [Tree](#)

Homo sapiens (29781320)
synthetic construct (57325)
Mus musculus (3176)
Human papillomavirus 16 (78)
Trypanosoma cruzi (21)
All other taxa (78)
[More...](#)

Find related data

Database: Select

[Search details](#)

in the nucleotide database in genbank it shows **29,841,998**

EMBL

this link

ebi.ac.uk/ena/browser/advanced-search

Relaunch to update

Data Type Query Inclusion/Exclusion Fields Data Filters Results

Download ENA records: FASTA Download report: JSON

TEXT TSV

Accession	Description
A00383	H.sapiens NANB-hepatitis associated DNA
A06851	H.sapiens gene for relaxin, exon 1
A08002	Nucleotide sequence 7 from patent number EP0370719
A08905	H.sapiens (haplotype 2B, allele MS32, isolate English, serial number 31) minisatellite sequence
A11954	H.sapiens mRNA fragment for calcitonin
A14571	(2'-5') oligo A synthetase cDNA for 1.6kb RNA (E16,clone 9-21)
A25971	H.sapiens mRNA for T-cell receptor IGR b 20 Vbeta 9
A26249	Human shIL5R-alpha gene
A26437	Human granzyme B
A28104	Human GABAA receptor alpha-5 subunit

Items per page: 10 1 - 10 of 10000 (29244316 total)

New Search Back Copy Curl Request Copy Http Request

so **29,244,316**

finally

DDBJ = 35,116,401

GenBank = 29,841,998

EMBL = 29,224,316

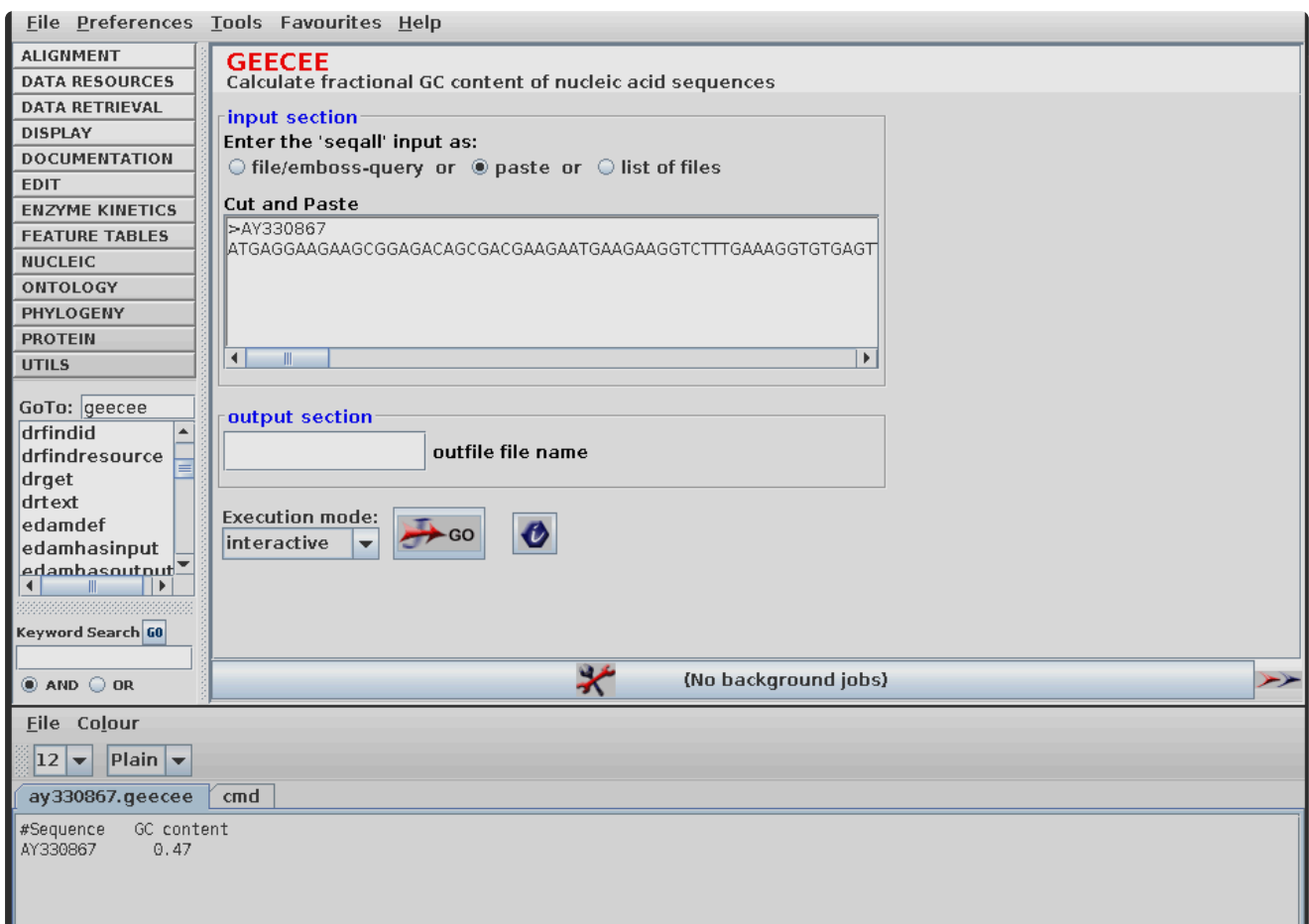
Q.2

link

this is the sequence taken from genbank

```
ATGAGGAAGAAGCGGAGACAGCGACGAAGAATGAAGAAGGTCTTTGAAAGGTGTGA
GTTGGCCAGAACTC
TGAAAAGATTGGGAATGGATGGCTACAGGGGAATCAGCCTAGCAAACCTGGATGTGT
TTGGCCAAATGGGA
GAGTGGTTACAACACACGAGCTACAACTACAATGCTGGAGACAGAAGCACTGATT
ATGGGATATTTTCAG
ATCAATAGCCGCTACTGGTGTAAATGATGGCAAAACCCCAGGAGCAGTTAATGCCTG
TCATTTATCCTGCA
GTGCTTTGCTGCAAGATAACATCGCTGATGCTGTAGCTTGTGCAAAGAGGGTTGTC
CGTGATCCACAAGG
CATTAGAGCATGGGTGGCATGGAGAAATCGTTGTCAAACAGAGGTGTCCGTCAGT
ATGTTCAAGGTTGT
GGAGTGTAAATGA
```

using emboss we can figure out the GC content



so it is 47%

Q.3

- <https://www.ncbi.nlm.nih.gov/nuccore/33088589>

- https://getentry.ddbj.nig.ac.jp/getentry/na/AY330867/?format=flatfile&filetype=html&trace=true&show_suppressed=false&limit=10
- <https://www.ebi.ac.uk/ena/browser/view/AY330867?show=orcid-links>

all databases give the sequence, translation and other useful info
all three are part of the INSDC and sync daily, so the core biological sequences and accession numbers are identical across all of them
(taken from ddbj)

LOCUS	AY330867	432 bp	mRNA	linear	SYN 29-JUL-2004
DEFINITION	Synthetic construct human lysozyme mRNA, complete cds.				
ACCESSION	AY330867				
VERSION	AY330867.1				
KEYWORDS	.				
SOURCE	synthetic construct				
ORGANISM	synthetic construct				
	other sequences; artificial sequences.				
REFERENCE	1 (bases 1 to 432)				
AUTHORS	Xie,X.D., Chen,Z.H. and Zhou,C.S.				
TITLE	Direct Submission				
JOURNAL	Submitted (26-JUN-2003) Postdoctoral Scientific Research Station of Gansu Yasheng Industrial (Group) Co., Ltd., Zhangye Road, Lanzhou, Gansu 730030, P.R.China				
FEATURES	Location/Qualifiers				
source	1..432				
	/organism="synthetic construct"				
	/mol_type="mRNA"				
	/db_xref="taxon:32630"				
CDS	1..429				
	/codon_start=1				
	/transl_table=11				
	/product="human lysozyme"				
	/protein_id="AAP93336.1"				
	/translation="MRKKRRQRRRMKKVFERCELARTLKRLGMDGYRGISLANWMCLA KWESGYNTRATNYNAGDRSTDYGIQINSRYWCNDGKTPGAVNACHLSCSALLQDNIA DAVACAKRVVRDPQGIRAWVAWRNRCQNRGVRQYVQGGCV"				
BASE COUNT	131 a	72 c	129 g	100 t	
ORIGIN	1 atgaggaaga agcggagaca gcgacgaaga atgaagaagg tctttgaaag gtgtgagttg				
	61 gccagaactc tgaaaagatt gggaatggat ggctacaggg gaatcagcct agcaaactgg				
	121 atgtgttttg ccaaatggga gagtgggttac aacacacgag ctacaaacta caatgctgga				
	181 gacagaagca ctgattatgg gatatttcag atcaatagcc gctactggtg taatgatggc				
	241 aaaacccag gagcagttaa tgcctgtcat ttatcctgca gtgctttgct gcaagataac				
	301 atcgctgat ctgtagcttg tgcaagagg gttgtccgtg atccacaagg cattagagca				
	361 tgggtggcat ggagaaatcg ttgtcaaaac agaggtgtcc gtcagtatgt tcaaggttgt				
	421 ggagtgtaat ga				
//					

but they have some differences like

- genbank and ddbj use full words like locus and origin for headers but embl uses two letter codes like id and sq
- ddbj clearly shows the exact base count right before the sequence but genbank skips it completely

- embl web view gives extra stuff like orcid data claims and cross references to other databases
- genbank and ddbj put the sequence numbers on the left side but embl puts them on the far right
- **genbank** it is tied to NCBI, meaning it natively cross-links to PubMed for literature, BLAST for alignments, and RefSeq for curated genomes
- **embl** it is built into the european framework, directly linking nucleotide sequences to Ensembl (genome browsers) and UniProt (protein databases)
- **ddbj** it serves the asian research node and links out to specific functional databases like KEGG for metabolic pathway analysis

the raw sequence data is exactly the same; the real difference is just the web interface and the downstream analytical tools each database connects us to

Q.4

[papers link](#)

The screenshot shows a Google Scholar search results page. The search query is "discrimination of beta barrel membrane proteins". The results are sorted by "YEAR". There are about 20,500 results in 0.41 seconds. The first six results are visible, each with a title, authors, year, journal, and a brief abstract. Each result also includes links for "Save", "Cite", "Cited by", "Related articles", and "All versions".

Title	Authors	Year	Journal	Source
A Hidden Markov Model method, capable of predicting and discriminating β -barrel outer membrane proteins	PG Bagos, TD Liakopoulos, IC Spyropoulos...	2004	BMC ...	Springer
TMBETADISC-RBF: discrimination of β -barrel membrane proteins using RBF networks and PSSM profiles	YY Ou, MM Gromiha, SA Chen, M Suwa	2008	Computational biology and ...	Elsevier
Predicting transmembrane beta-barrels in proteomes	HR Bigelow, DS Petrey, J Liu, D Przybylski...	2004	Nucleic acids ...	academic.oup.com
PRED-TMBB2: improved topology prediction and detection of beta-barrel outer membrane proteins	KD Tsirigos, A Elofsson, PG Bagos	2016	Bioinformatics	academic.oup.com
A sequence-profile-based HMM for predicting and discriminating beta barrel membrane proteins	PL Martelli, P Fariselli, A Krogh, R Casadio	2002	ISMB	academia.edu
Functional discrimination of membrane proteins using machine learning techniques	MM Gromiha, Y Yabuki	2008	BMC bioinformatics	Springer

some examples:

Liu, Q., Zhu, Y., Wang, B., & Li, Y. (2003). Identification of β -barrel membrane proteins based on amino acid composition properties and predicted secondary structure. *Computational Biology and Chemistry*, 27, 355–361. [https://doi.org/10.1016/s1476-9271\(02\)00085-3](https://doi.org/10.1016/s1476-9271(02)00085-3) Cited by: 52

Ou, Y.-Y., Gromiha, M. M., Chen, S.-A., & Suwa, M. (2008). TMBETADISC-RBF: Discrimination of β -barrel membrane proteins using RBF networks and PSSM profiles. *Computational Biology and*

Chemistry, 32, 227–231.

<https://doi.org/10.1016/j.compbiolchem.2008.03.002> Cited by: 106

Pubmed link as well

The screenshot shows the PubMed website interface. At the top, there's a header for the NIH National Library of Medicine. Below that, the PubMed logo is visible. A search bar contains the query "discrimination of beta barrel membrane proteins". To the right of the search bar are buttons for "Advanced", "Create alert", "Create RSS", and "Search". Below the search bar, there are buttons for "Save", "Email", and "Send to". To the right of these buttons are "Sort by: Best match" and "Display options". On the left side, there's a section for "MY CUSTOM FILTERS" with a link to "Edit custom filters". Below that is a "RESULTS BY YEAR" bar chart showing a distribution of results from 1992 to 2026. Further down are "PUBLICATION DATE" filters for 1 year, 5 years, 10 years, and Custom Range. At the bottom left is a "TEXT AVAILABILITY" section with checkboxes for "Abstract" and "Free full text". The main results area shows "54 results" and a pagination bar indicating "Page 1 of 6". A blue banner states "2 articles found by citation matching". Below this, two articles are listed: "TMBETADISC-RBF: Discrimination of beta-barrel membrane proteins using RBF networks and PSSM profiles." by Ou YY, et al. (2008, PMID: 18434251) and "Current developments on beta-barrel membrane proteins: sequence and structure analysis, discrimination and prediction." by Gromiha MM, et al. (2007, PMID: 18220845). A yellow warning box states "Quoted phrase not found in phrase index: 'discrimination of beta barrel membrane proteins'". At the bottom, a link to "OMPdb: A Global Hub of Beta-Barrel Outer Membrane Proteins." is shown, along with a citation for Roumia AF, et al. (2021, PMID: 36303794) and a link to the "Free PMC article".

Tsirigos, K. D., Elofsson, A., & Bagos, P. G. (2016). PRED-TMBB2: improved topology prediction and detection of beta-barrel outer membrane proteins. *Bioinformatics*, 32, i665–i671.

<https://doi.org/10.1093/bioinformatics/btw444> Cited by: 103

Q.5

Kihara D Google Scholar link

one paper published in nature journal example

paper link

they report the results from the first large-scale community-based critical assessment of protein function annotation (CAFA) experiment.

on pubmed

 An official website of the United States government [Here's how you know](#) ✓

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User Guide

Add terms to the query box

Author

⬆ ⬇ ⬆

Kihara D.

✕

ADD ▾

Show Index

Query box

Enter / edit your search query here

⌵

Search ▾

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Kihara D.[Author] × **Search**

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[Filters](#) [Timeline](#) Sort by: [Best match](#) [Display options](#)

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275 results « < Page 1 of 28 > »

i It looks like you are searching for an author. Results are currently sorted by Best Match. ×
To see the newest results first, [change the sort order to Most Recent](#).

☐ [Effect of phosphorylation barcodes on arrestin binding to a chemokine receptor.](#)
1 Chen Q, Schafer CT, Mukherjee S, Wang K, Gustavsson M, Fuller JR, Tepper K, Lamme TD, Aydin Y,
Cite Agrawal P, Terashi G, Yao XQ, **Kihara D**, Kosiakoff AA, Handel TM, Tesmer JG.
Nature. 2025 Jul;643(8070):280-287. doi: 10.1038/s41586-025-09024-9. Epub 2025 May 21.
PMID: 40399676 [Free PMC article](#).

☐ [Computational protein function predictions.](#)
2 **Kihara D**.
Cite Methods. 2016 Jan 15;93:1-2. doi: 10.1016/j.ymeth.2016.01.001.
PMID: 26778120 No abstract available.

☐ [Computational prediction of disordered binding regions.](#)
3 **Kihara D**, **Kihara D**, **Kihara D**.

Q.6

paper link

"Mitochondrial beta-barrel proteins, an exclusive club?"

there are lots of related articles



Cell 2008 Dec 26;135(7):1158-9?



Search

[Advanced](#) [Create alert](#) [Create RSS](#)

[User Guide](#)

Found 1 result for an alternative search.

Your search for *Cell 2008 Dec 26;135(7):1158-9?* retrie...

Save

Email

Send to

Display options 

[Comment](#) > [Cell. 2008 Dec 26;135\(7\):1158-9; author reply 1159-60.](#)

doi: 10.1016/j.cell.2008.12.017.

Mitochondrial beta-barrel proteins, an exclusive club?

[Kenichiro Imai](#), [M Michael Gromiha](#), [Paul Horton](#)

PMID: 19109882 DOI: [10.1016/j.cell.2008.12.017](#) 

Free article

 Full text links

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No abstract available

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Comment on

Similar articles

[Transport proteins \(carriers\) of mitochondria.](#)

Wohlrab H.

IUBMB Life. 2009 Jan;61(1):40-6. doi: 10.1002/iub.139.

PMID: 18816452 Review.

[Systematic analysis of the twin cx\(9\)c protein family.](#)

Longen S, Bien M, Bihlmaier K, Kloeppel C, Kauff F, Hammermeister M, Westermann B, Herrmann JM, Riemer J.

J Mol Biol. 2009 Oct 23;393(2):356-68. doi: 10.1016/j.jmb.2009.08.041. Epub 2009 Aug 21.

PMID: 19703468

[Mitochondrial permeability transition pore opening as a promising therapeutic target in cardiac diseases.](#)

Javadov S, Karmazyn M, Escobales N.

J Pharmacol Exp Ther. 2009 Sep;330(3):670-8. doi: 10.1124/jpet.109.153213. Epub 2009 Jun 9.

PMID: 19509316

[Vitamin E protects against the mitochondrial damage caused by cyclosporin A in LLC-PK1 cells.](#)

de Arriba G, de Hornedo JP, Rubio SR, Fernández MC, Martínez SB, Camarero MM, Cid TP.

Toxicol Appl Pharmacol. 2009 Sep 15;239(3):241-50. doi: 10.1016/j.taap.2009.05.028. Epub 2009 Jun 11.

PMID: 19523970

[A CaPful of mechanisms regulating the mitochondrial permeability transition.](#)

Di Lisa F, Bernardi P.

J Mol Cell Cardiol. 2009 Jun;46(6):775-80. doi: 10.1016/j.yjmcc.2009.03.006. Epub 2009 Mar 19.

PMID: 19303419 Review.

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Total 27 similar articles are listed [here](#)

on pubmed as well

Comment on

Full text links

Cell. 2008 Mar 21;132(6):1011-24. doi: 10.1016/j.cell.2008.01.028.
PMID: 18358813

Similar articles

[Transport proteins \(carriers\) of mitochondria.](#)
Wohlrab H.
IUBMB Life. 2009 Jan;61(1):40-6. doi: 10.1002/iub.139.
PMID: 18816452 Review.

[Systematic analysis of the twin cx\(9\)c protein family.](#)
Longen S, Bien M, Bihlmaier K, Kloeppel C, Kauff F, Hammermeister M, Westermann B, Herrmann JM, Riemer J.
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PMID: 19703468

[Mitochondrial permeability transition pore opening as a promising therapeutic target in cardiac diseases.](#)
Javadov S, Karmazyn M, Escobales N.
J Pharmacol Exp Ther. 2009 Sep;330(3):670-8. doi: 10.1124/jpet.109.153213. Epub 2009 Jun 9.
PMID: 19509316

[Vitamin E protects against the mitochondrial damage caused by cyclosporin A in LLC-PK1 cells.](#)
de Arriba G, de Hornedo JP, Rubio SR, Fernández MC, Martínez SB, Camarero MM, Cid TP.
Toxicol Appl Pharmacol. 2009 Sep 15;239(3):241-50. doi: 10.1016/j.taap.2009.05.028. Epub 2009 Jun 11.
PMID: 19523970

[A CaPful of mechanisms regulating the mitochondrial permeability transition.](#)
Di Lisa F, Bernardi P.
J Mol Cell Cardiol. 2009 Jun;46(6):775-80. doi: 10.1016/j.yjmcc.2009.03.006. Epub 2009 Mar 19.
PMID: 19303419 Review.

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Q.7

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National Center for Biotechnology Information

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PUBLICATION DATE

☒ 1 year
☐ 5 years
☐ 10 years
☐ Custom Range

TEXT AVAILABILITY

☐ Abstract
☐ Free full text
☐ Full text

ARTICLE ATTRIBUTE

2,937 results

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- ☐ **Ketogenic diet mediates intestinal tumorigenesis through lipids not ketones.**
 1 Shay JES, Chi F, Tzouanas CN, Han S, Zhang X, Ten Hoeve J, Williams KJ, Neptun S, Sever T, Fuentes I, Bhatia SN, Calibasi-Kocal G, Vander Heiden MG, Shalek AK, Yilmaz ÖH.
 Cite **Nature.** 2026 Aug;656(8127):474-483. doi: 10.1038/s41586-026-10779-y. Epub 2026 Jul 15. PMID: 42457955
- ☐ **GPC3-specific dnTGFβRII-armoured CAR T cells for hepatocellular carcinoma.**
 2 Zhang Q, Fu Q, Shen Y, Cao W, Jin G, Zhang Y, Wu J, Chen C, Wang H, Xu X, Sun K, Xue X, Bondanza A, Cao D, Chu N, Durham N, Cobbold M, Farsaci B, Giardino Torchia ML, Hao Y, Hong Y, Huang J, Humphries M, Jian Q, Moody G, Stone J, Sun L, Tu E, Wang F, Wang F, Yang Y, Yao Y, Zou A, Bai X, Liang T.
 Cite **Nature.** 2026 Jul 15. doi: 10.1038/s41586-026-10786-z. Online ahead of print. PMID: 42457964
- ☐ **Diet-microbiome synergy underlies obesity-associated immunotherapy efficacy.**
 3 Desharnais L, Swaby A, Messaoudene M, Doré S, Yu MW, Fiset B, Breton V, Ponce M, Hu Y, Wilson L, Sorin M, Wang Y, Dewar K, Pollak M, Elkrief A, Routy B, Walsh LA, Quail DF.
 Cite **Nature.** 2026 Jul 8. doi: 10.1038/s41586-026-10750-x. Online ahead of print. PMID: 42420462
- ☐ **Mitochondrial metabolism and epigenetic crosstalk drive SASP.**
 4 Martini H, Birch J, Marques EDM, Vittorelli S, Lannado AB, Pirius N, Franco AC, Lee G, Han Y, Rowsay II

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☐ All ☐ Export Download Citation overview ... More Show all abstracts Sort by Date (newest) ☐ ☐

	Document title	Authors	Source	Year	Citations
☐ 1	Article • Open access Minimalist terahertz wireless transceiver in integrated photonics	Guo Y, Zhang X, Zhang Y, ... Wang X, Shu H.	Nature Communications , 17(1), 4429	2026	0
	Show abstract Full Text View at Publisher Related documents				
☐ 2	Article • Open access CXCR3 is associated with T-cell-induced heart damage in acute rheumatic fever	Middleton F.M., McGregor R., Lorenz N., ... Carapetis J., Moreland N.J.	Nature Communications , 17(1), 4664	2026	0
	Show abstract Full Text View at Publisher Related documents				

SCOPUS shows 22,924 articles

Q.8

Burkhard Rost [FOLLOW](#) [GET MY OWN PROFILE](#)

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[machine learning](#) [artificial intelligence](#) [bioinformatics](#) [computer science](#) [protein prediction](#)

TITLE	CITED BY	YEAR
The transcriptional landscape of the mammalian genome P Carninci, T Kasukawa, S Katayama, J Gough, MC Frith, N Maeda, ... science 309 (5740), 1559-1563	4251	2005
Prediction of protein secondary structure at better than 70% accuracy B Rost, C Sander Journal of molecular biology 232 (2), 584-599	3970	1993
ProtTrans: toward understanding the language of life through self-supervised learning	3809 *	2021

Cited by [VIEW ALL](#)

	All	Since 2021
Citations	65146	19704
h-index	116	60
i10-index	292	182

Bar chart showing citation trends over time (2005-2021).

on google scholar

Institutional info

Scopus

This author profile is generated by Scopus 21

Rost, Burkhard

Technische Universität München, Munich, Germany • Scopus ID: 5525157100 • 0000-0002-0179-8624 7

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Article • Open access
BioCentral: Embedding-based Protein Predictions
Frans, Sebastian, Olery, Tobias, Schaefermann, Pauls, ... Jiménez-Goto, Luisa F., Rost, Burkhard
Journal of Molecular Biology, 2026, 418(18), 169673
Show abstract Full text

Article • Open access
Whole-genome prediction of bacterial pathogenic capacity on novel bacteria

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on pubmed

41,446 citations and 96 h-index

Q.9

link to brenda

Information on EC 1.7.2.3 - trimethylamine-N-oxide reductase

for references in articles please use BRENDA:EC1.7.2.3

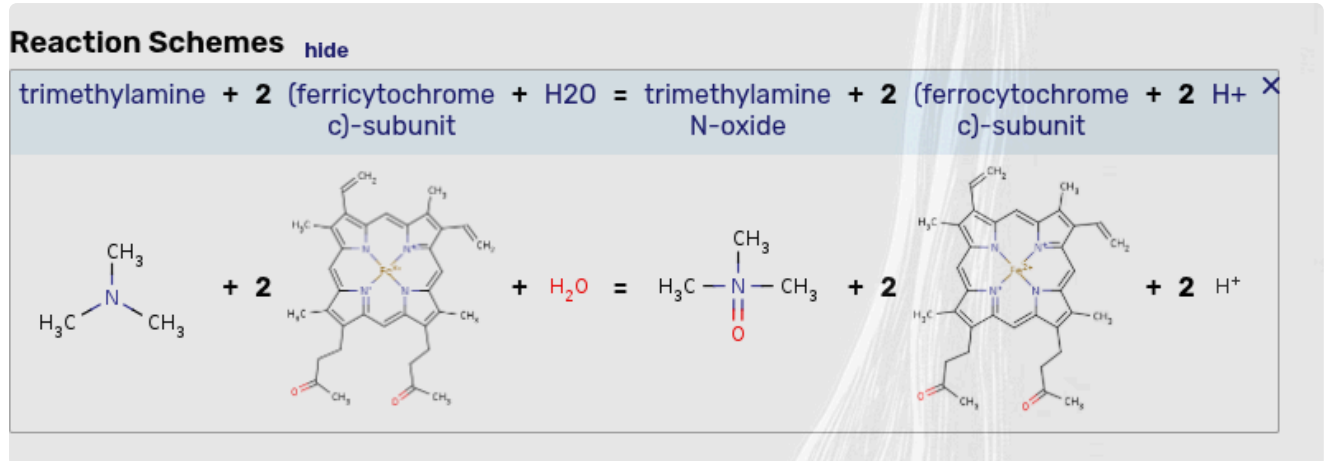
EC Tree → 1 Oxidoreductases → 1.7 Acting on other nitrogenous compounds as donors → 1.7.2 With a cytochrome as acceptor → 1.7.2.3 trimethylamine-N-oxide reductase

this shows the enzyme class hierarchy

- **Accepted Name:** Trimethylamine-N-oxide reductase (cytochrome c), also commonly known as TMAO reductase.
- **Enzyme Class Hierarchy:**
 - **EC 1 (Class):** Oxidoreductases (enzymes that catalyze oxidation-reduction reactions).
 - **EC 1.7 (Subclass):** Acting on other nitrogenous compounds as donors.
 - **EC 1.7.2 (Sub-subclass):** With a cytochrome as an acceptor.

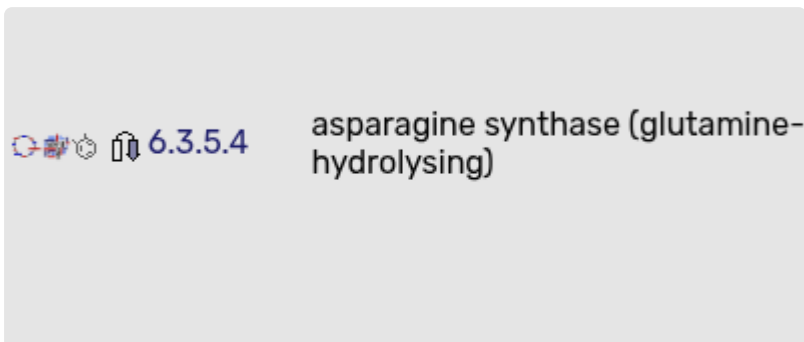
- **Function / Catalytic Activity:** This microbial enzyme catalyzes the chemical reduction of trimethylamine N-oxide (TMAO) into trimethylamine (TMA) as part of the electron transport chain.

reaction



Q.10

finding the EC number on brenda



now searching it on ebi(thornton) gives us the enzyme

<https://www.ebi.ac.uk/thornton-srv/m-csa/entry/302/>

[1ct9Q](#) - CRYSTAL STRUCTURE OF ASPARAGINE SYNTHETASE B FROM ESCHERICHIA COLI (2.0 Å)  

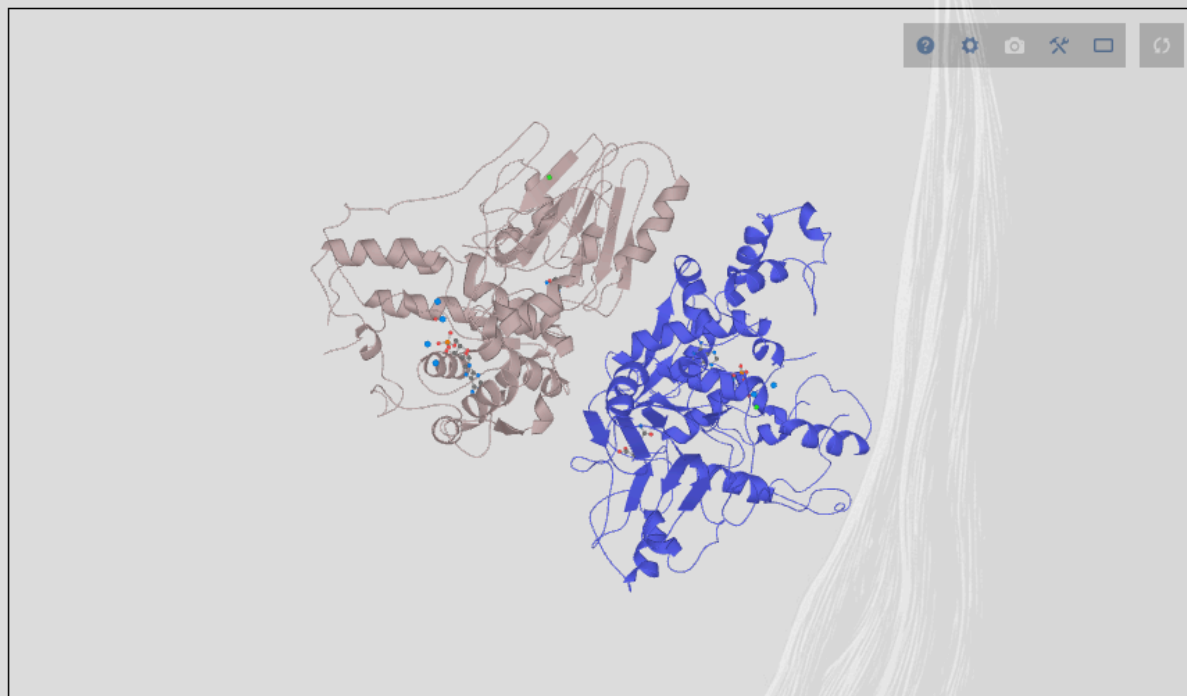


Catalytic CATH Domains

[3.60.20.10Q](#)  [3.40.50.620Q](#)  (see all for [1ct9Q](#))

Cofactors

Magnesium(2+) (1) [Metal MACiE](#)



Show Active Center

and below it shows the catalytic residues and their roles

Catalytic Residues Roles			
UniProt	PDB* (1ct9)		
Cys2 (N-term)	Ala1A (N-term)	Acts as a general acid/base to activate the cysteine nucleophile.	proton acceptor, proton donor
Leu51 (main-C)	Leu50A (main-C)	Helps stabilise the reactive intermediates formed.	hydrogen bond acceptor, electrostatic stabiliser
Thr322, Arg325	Thr321A, Arg324A	Bind and stabilise the phosphate groups of the ATP and reactive intermediates formed.	hydrogen bond donor, electrostatic stabiliser
Cys2	Ala1A	Acts as a catalytic nucleophile in the glutaminase domain reaction.	covalently attached, hydrogen bond acceptor, nucleofuge, nucleophile, proton acceptor, proton donor
Gly76 (main-N), Asn75	Gly75A (main-N), Asn74A	Forms the oxyanion hole that stabilises the reactive intermediates and transition states formed.	hydrogen bond donor, electrostatic stabiliser

so answer = Cys 1, Cys 2, Gly76, Leu 51, Thr 322, and Arg 325.

Q.11

Search for as complete name

Homo sapiens

Taxonomy ID: 9606 (for references in articles please use ncbitaxon:9606)

current name

Homo sapiens Linnaeus, 1758

Genbank common name: **human**

NCBI BLAST name: **primates**

Rank: **species**

Genetic code: [Translation table 1 \(Standard\)](#)

Mitochondrial genetic code: [Translation table 2 \(Vertebrate Mitochondrial\)](#)

Lineage(full)

[cellular organisms](#); [Eukaryota](#); [Opisthokonta](#); [Metazoa](#); [Eumetazoa](#); [Bilateria](#); [Deuterostomia](#); [Chordata](#); [Craniata](#); [Vertebrata](#); [Gnathostomata](#); [Teleostomi](#); [Euteleostomi](#); [Superstomi](#)

Felis catus

Taxonomy ID: 9685 (for references in articles please use ncbitaxon:9685)

current name

Felis catus Linnaeus, 1758

includes: *Korat cats* L.

homotypic synonym:
Felis silvestris catus

Genbank common name: domestic cat

just listing the data below now

Organism	Scientific Name	Taxonomy ID	Number of Chromosomes (2n)
Human	<i>Homo sapiens</i>	9606	46
Cat	<i>Felis catus</i>	9685	38
Dog	<i>Canis lupus familiaris</i>	9615	78
Domestic guinea pig	<i>Cavia porcellus</i>	10141	64
Thale cress	<i>Arabidopsis thaliana</i>	3702	10

Q.12

The **Entrez Programming Utilities (E-utilities)** are a suite of server-side programs that serve as the public API for the NCBI Entrez system. They allow users to programmatically search, link, and retrieve data from across all NCBI Entrez databases (including PubMed, Nucleotide, Protein, and Gene) without needing to use the web browser interface.

example syntax

```
esearch.fcgi?db=<database>&term=<query>
```

example

- `[https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esearch.fcgi?db=pubmed&term=science[journal]+AND+breast+cancer+AND+2008[pdat]]`
`(https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esearch.fcgi?db=pubmed&term=science%5bjournal%5d+AND+breast+cancer+AND+2008%5bpdat%5d)`

to fetch a fasta format

```
[https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?db=nuccore&id=34577062,24475906&rettype=fasta&retmode=text]
```

syntax is

- `db=nuccore`: Specifies the database (e.g., `nuccore` for nucleotide, or `protein`).
- `id=AY330867`: The unique identifier or accession number of the record.
- `rettype=fasta`: Tells the server you want the specific FASTA layout.
- `retmode=text`: Tells the server to return plain text rather than XML.

Q.13

a. Protein properties

NAR Database Summary Paper Category List

Nucleotide Sequence Databases
 RNA sequence databases
 Protein sequence databases
 Structure Databases
 Genomics Databases (non-vertebrate)
 Metabolic and Signaling Pathways
 Human and other Vertebrate Genomes
 Human Genes and Diseases
 Microarray Data and other Gene Expression Databases
 Proteomics Resources
 Other Molecular Biology Databases
 Organelle databases
 Plant databases
 Immunological databases
 Cell biology

► Compilation Paper
 ► Category List
 ► Alphabetical List
 ► Category/Paper List
 ► Search Summary Papers

► Compilation Paper
 ► Category List
 ► Alphabetical List
 ► Category/Paper List
 ► Search Summary Papers

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General sequence databases

Protein properties

AAindex
 BindingDB
 Cybase
 dbPTM
 eProS
 HHMD
 IProLINK
 MALISAM
 MegaMotifbase
 MemMoRF
 Minimotif miner
 MobiDB
 MP:PD
 PFD - Protein Folding Database
 PIDD
 PINT
 PPD
 PPT-DB
 Proteome-pI 2.0
 ProTherm
 PRTAD
 REFOLD
 SwissSidechain
 TOPPR

1. **UniProtKB (Universal Protein Resource):** The central hub for the collection of functional information on proteins, with accurate, consistent, and rich annotation of protein properties.
2. **Pfam (or InterPro):** A large collection of protein families, represented by multiple sequence alignments and hidden Markov models (HMMs), used to determine protein domain properties.

**b. Small molecules (Structure related)*

Transcriptional regulator sites and transcription facto

Small molecules

BitterDB
ChEBI - Chemical Entities of Biological Interest
ChemBank
ChemDB
Crystallography Open Database
CSD - Cambridge Structural Database
EDULISS (currently unavailable)
FragmentStore
Hemolytik
Het-PDB Navi
HIC-Up
MMsINC
mVOC
PDB-Ligand
PubChem
R.E.DD.B.
SCRIPDB
SuperDrug
SuperScent
SuperToxic

3. **PubChem:** A massive database maintained by NCBI that provides information on the chemical structures, physical properties, and biological activities of small molecules
4. **ChEBI (Chemical Entities of Biological Interest):** A freely available dictionary of molecular entities focused on strictly "small" chemical compounds and their structural properties.

c. Cancer gene databases

Cancer gene databases

ArrayMap
BCCTBbp
BreCAN-DB
Cancer RNA-Seq Nexus
Cancer3D
CancerGenes
CancerPPD
Candidate Cancer Gene Database
CanGEM
CanSAR
CaSNP
cBioPortal
CCDB
ccmGDB
CellLineNavigator
ChimerDB
CIVICdb
ClinVar
CMPD
Colorectal Cancer Atlas
COLT-Cancer
COSMIC
CTDatabase
dbDEMC
dbDEPC
DIVAS
DriverDBv2
FusionCancer
HLungDB
HPTAA
IARC TP53 Database
IGDB.NSCLC
intOGen
ITTACA
Inc2cancer
MethHC
MethyCancer
MoKCa
MutationAligner
Network of Cancer Genes
OncoDB.HCC
OpenTein
Pancreas Expression
PRECOG and related resources
rSNPBase 3.0

5. **COSMIC (Catalogue of Somatic Mutations in Cancer):** The world's largest and most comprehensive resource for exploring the impact of somatic mutations in human cancer.
6. **cBioPortal for Cancer Genomics:** An open-access resource for interactive exploration of multidimensional cancer genomics data sets.

Class Assignment

Question 1

Sequences:

- **Sequence 1:** ACAGACGTCCG (11 bases)
- **Sequence 2:** ACCGTCCG (8 bases)

Scoring Parameters:

- **Match:** +2
- **Mismatch:** -1
- **Gap:** -2

Recurrence Formulas:

- **Diagonal:** $F(i - 1, j - 1) + (+2 \text{ if match, } -1 \text{ if mismatch})$
- **Up (Gap in Sequence 2):** $F(i - 1, j) - 2$
- **Left (Gap in Sequence 1):** $F(i, j - 1) - 2$
- **Cell Score:** $F(i, j) = \max(\text{Diagonal, Up, Left})$

_	**_**	**A**	**C**	**C**	**G**	**T**	**C**	**C**	**G**
_	**0**	-2	-4	-6	-8	-10	-12	-14	-16
A	-2	**2**	0	-2	-4	-6	-8	-10	-12
C	-4	0	**4**	2	0	-2	-4	-6	-8
A	-6	-2	**2**	3	1	-1	-3	-5	-7
G	-8	-4	**0**	1	5	3	1	-1	-3
A	-10	-6	**2**	-1	3	4	2	0	-2
C	-12	-8	-4	**0**	1	2	6	4	2
G	-14	-10	-6	-2	**2**	0	4	5	6
T	-16	-12	-8	-4	0	**4**	2	3	4
C	-18	-14	-10	-6	-2	2	**6**	4	2
C	-20	-16	-12	-8	-4	0	4	**8**	6
G	-22	-18	-14	-10	-6	-2	2	6	**10**

Traceback:

Start from the bottom-right cell (10) and trace backward through the cells that yielded each optimal score until reaching the origin (0,0).

optimal global alignment

```

A C A G A C G T C C G
A C - - - C G T C C G

```

Question 2

The sequences are:

- Sequence 1: ACGTATCGCCGTATA (15 bases)
- Sequence 2: GATGCTCTCCGAAA (14 bases)

Scoring parameters

- Match: +2
- Mismatch: -1
- Gap: -2

For the Smith-Waterman formula,

0 is included as a lower bound because this is a local alignment. If continuing the alignment produces a negative score, the score resets to 0 and a new alignment can begin at that position:

$$H(i, j) = \max(0, H(i-1, j-1) + \text{score}, H(i-1, j) - 2, H(i, j-1) - 2)$$

** *	** *	**G**	**A**	**T**	**G**	**C**	**T**	**C**	**T**	**C**	**C**	**G**	**A**	**A**	**A**
-	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
A	0	0	2	0	0	0	0	0	0	0	0	0	2	2	2
C	0	0	0	1	0	2	0	2	0	2	2	0	0	1	1
G	0	2	0	0	3	1	1	0	1	0	1	4	2	0	0
T	0	0	1	2	1	2	3	1	2	0	0	2	3	1	0
A	0	0	2	0	1	0	1	2	0	1	0	0	4	5	3
T	0	0	0	4	2	0	2	0	4	2	0	0	2	3	4
C	0	0	0	2	3	4	2	4	2	6	4	2	0	1	2
G	0	2	0	0	4	2	3	2	3	4	5	6	4	2	0
C	0	0	1	0	2	6	4	5	3	5	6	4	5	3	1
C	0	0	0	0	0	4	5	6	4	5	7	5	3	4	2
G	0	2	0	0	2	2	3	4	5	3	5	9	7	5	3
T	0	0	1	2	0	1	4	2	6	4	3	7	8	6	4
A	0	0	2	0	1	0	2	3	4	5	3	5	9	10	8
T	0	0	0	4	2	0	2	1	5	3	4	3	7	8	9
A	0	0	2	2	3	1	0	1	3	4	2	3	5	9	10

For Needleman–Wunsch, the traceback starts at the bottom-right corner. For Smith–Waterman, traceback starts at the highest score anywhere in the table (10) and stops when a cell with a score of 0 is reached.

optimal local alignment

TCGCCGTATA

TCTCCG-AAA